

SQL

OLTP VS OLAP



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OLTP and OLAP are two types of data processing systems used in SQL databases



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OLTP

OLTP stands for Online Transaction Processing.



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OLTP systems are designed to **manage transaction-oriented** applications.

These applications are characterized by a large number of short online transactions (INSERT, UPDATE, DELETE).



OLTP systems are optimized for transactional throughput, ensuring data integrity, and handling a high volume of concurrent transactions.

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They typically have **normalized data schemas** to minimize redundancy and maintain data integrity.



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OLAP

OLAP stands for Online Analytical Processing



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OLAP systems are designed for **complex queries and analysis of data**, usually for decision-making purposes.

They are optimized for read-heavy operations and complex analytical queries.



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OLAP systems often involve data aggregation, such as summarization or consolidation of data across multiple dimensions.

They typically have **denormalized or star/snowflake schema designs**, optimized for query performance rather than transactional efficiency.



OLAP databases are commonly used in data warehousing environments, business intelligence applications, and reporting systems.



OLTP Example

When you **add an item to your cart** on an online store, that's an OLTP operation.



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OLAP Example

When a manager **checks the quarterly sales report** to see which products sold the most, that's an OLAP operation.



In Summary

OLTP systems are for handling **day-to-day transactional** operations.

OLAP systems are for **analyzing and gaining insights** from large volumes of data.

**I hope you find this
valuable! 🥰**

**A like from you would be a great
encouragement**



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